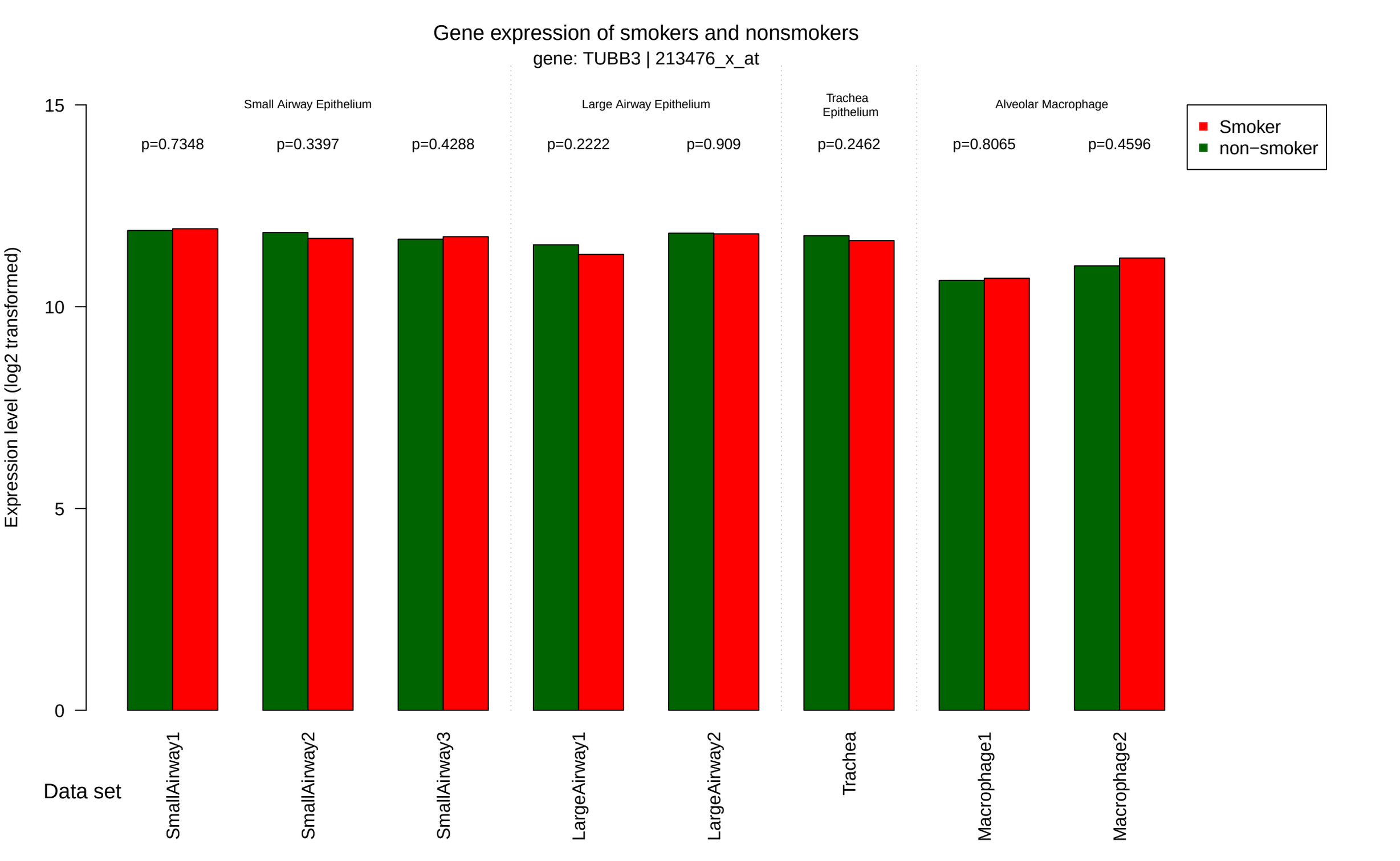
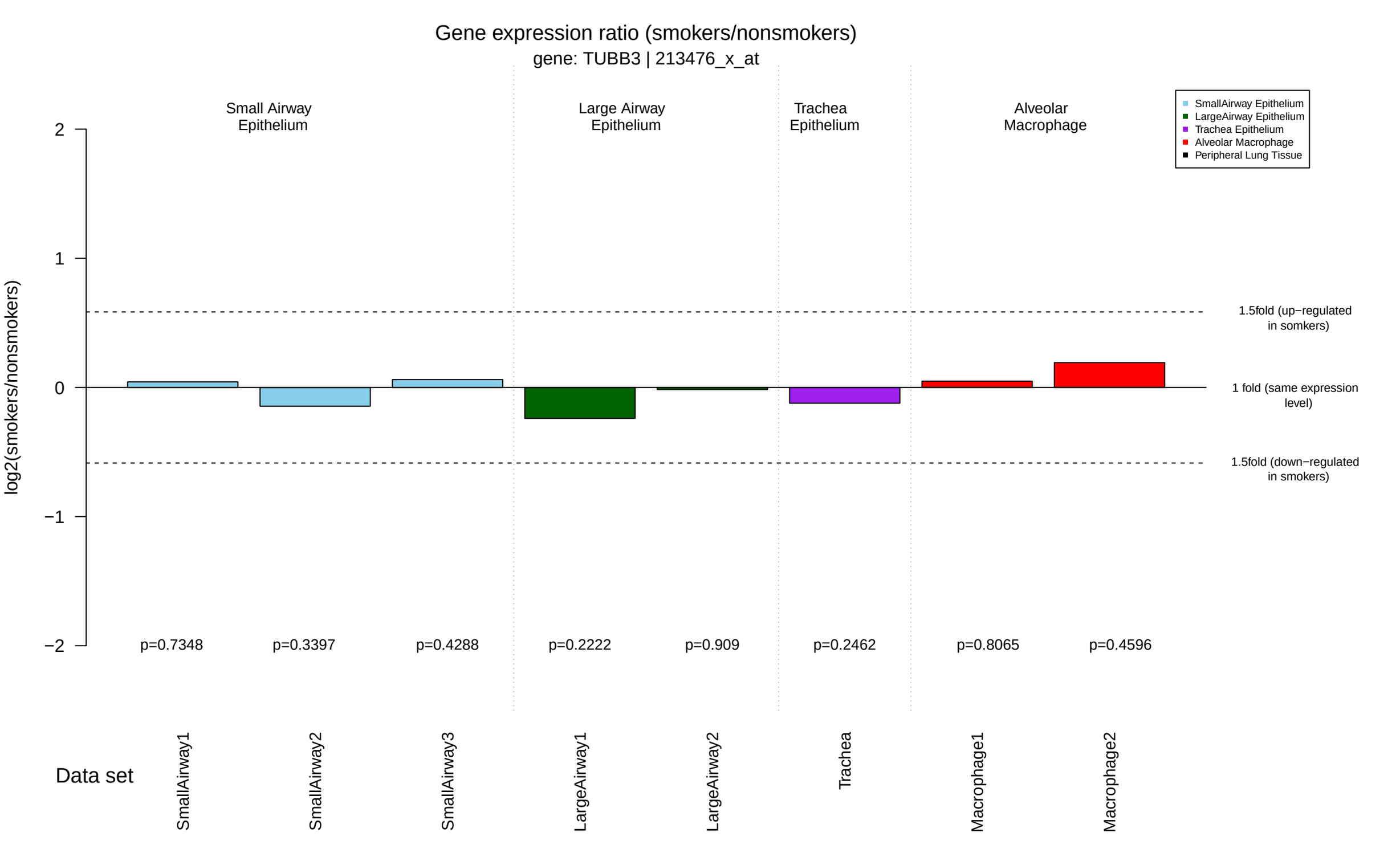
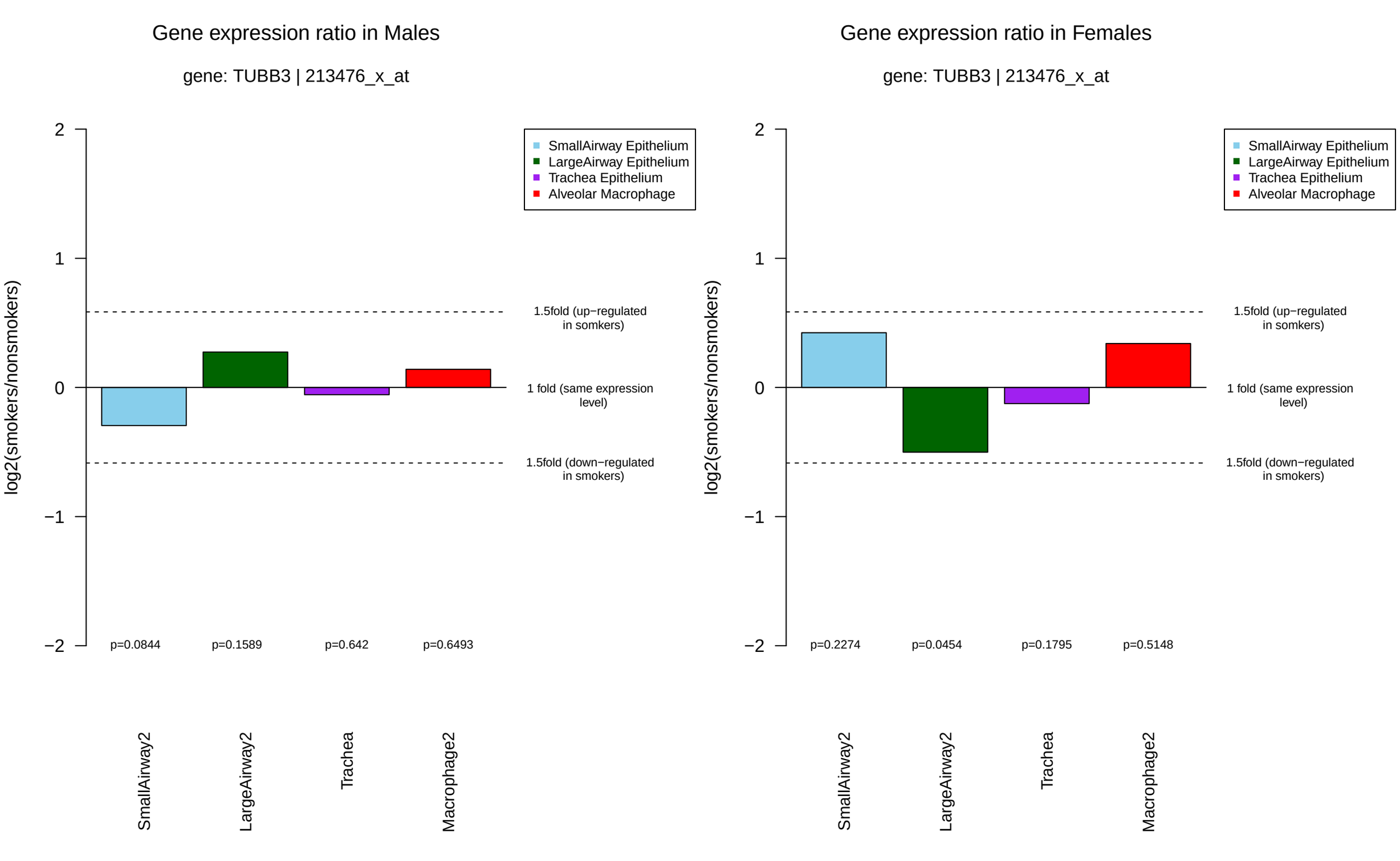
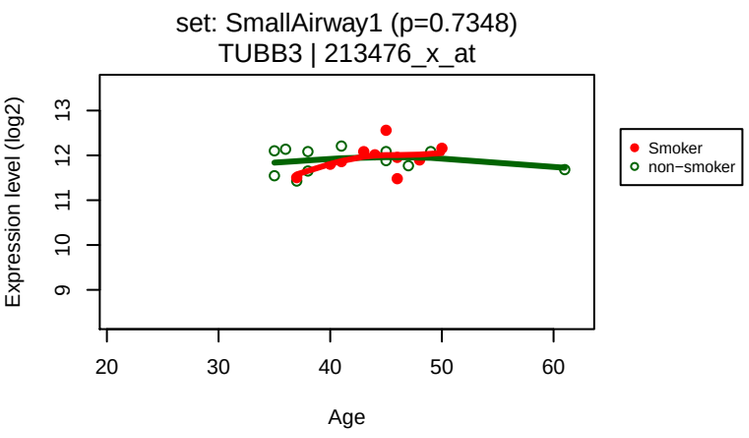
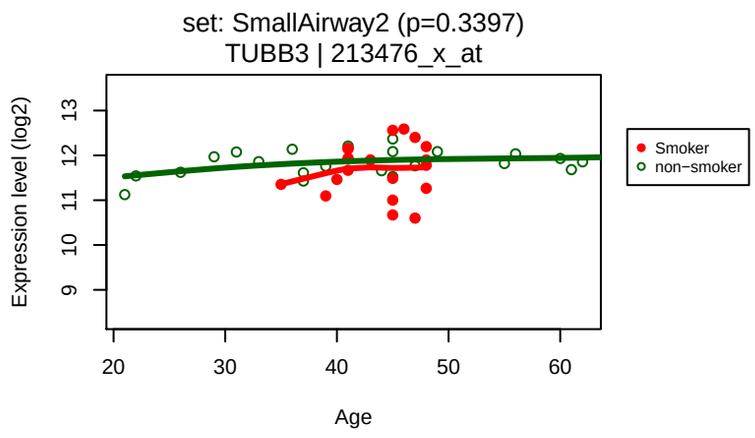
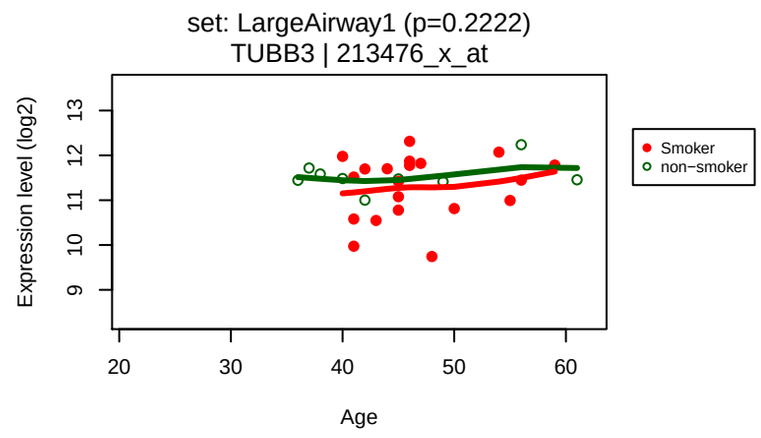

Gene expression of smokers and nonsmokers
gene: TUBB3 | 213476_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7348
p=0.3397
p=0.4288
p=0.2222
p=0.909
p=0.2462
p=0.8065
p=0.4596
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

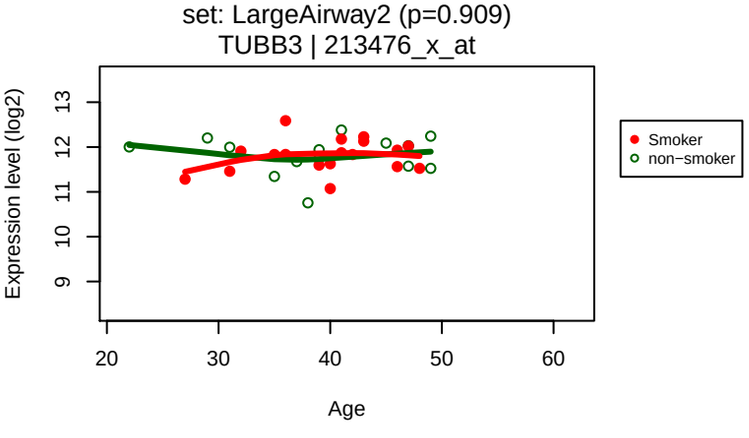

Gene expression ratio (smokers/nonsmokers)
gene: TUBB3 | 213476_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7348
p=0.3397
p=0.4288
p=0.2222
p=0.909
p=0.2462
p=0.8065
p=0.4596
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

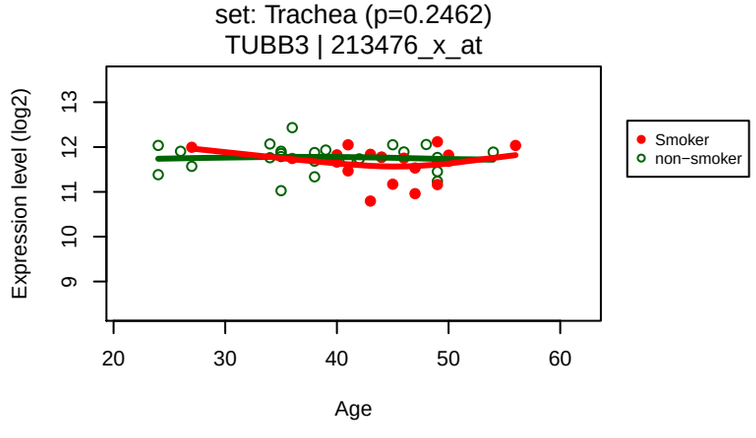

Gene expression ratio in Males
gene: TUBB3 | 213476_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0844
p=0.1589
p=0.642
p=0.6493
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TUBB3 | 213476_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2274
p=0.0454
p=0.1795
p=0.5148
SmallAirway2
LargeAirway2
Trachea
Macrophage2

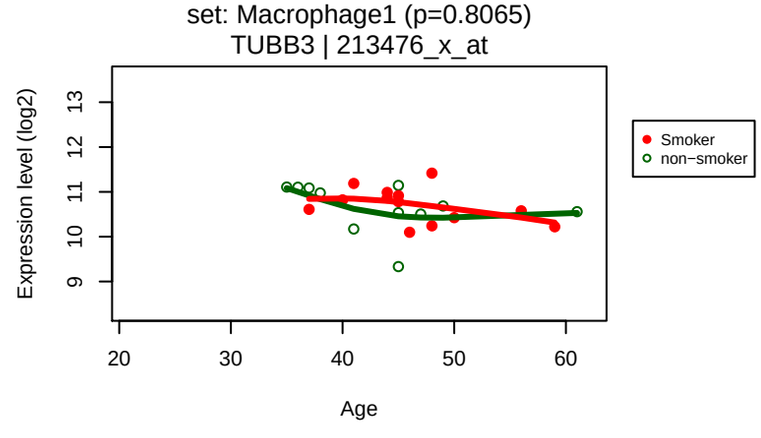

set: SmallAirway1 (p=0.7348)
TUBB3 | 213476_x_at
Expression level (log2)
Age
Smoker
non−smoker

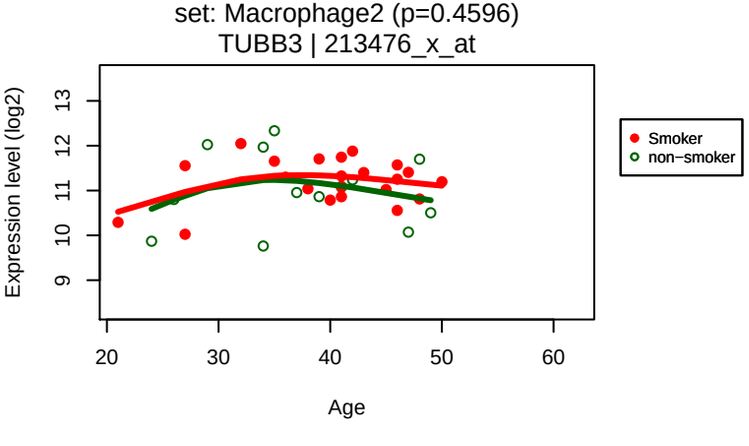

set: SmallAirway2 (p=0.3397)
TUBB3 | 213476_x_at
Expression level (log2)
Age
Smoker
non−smoker

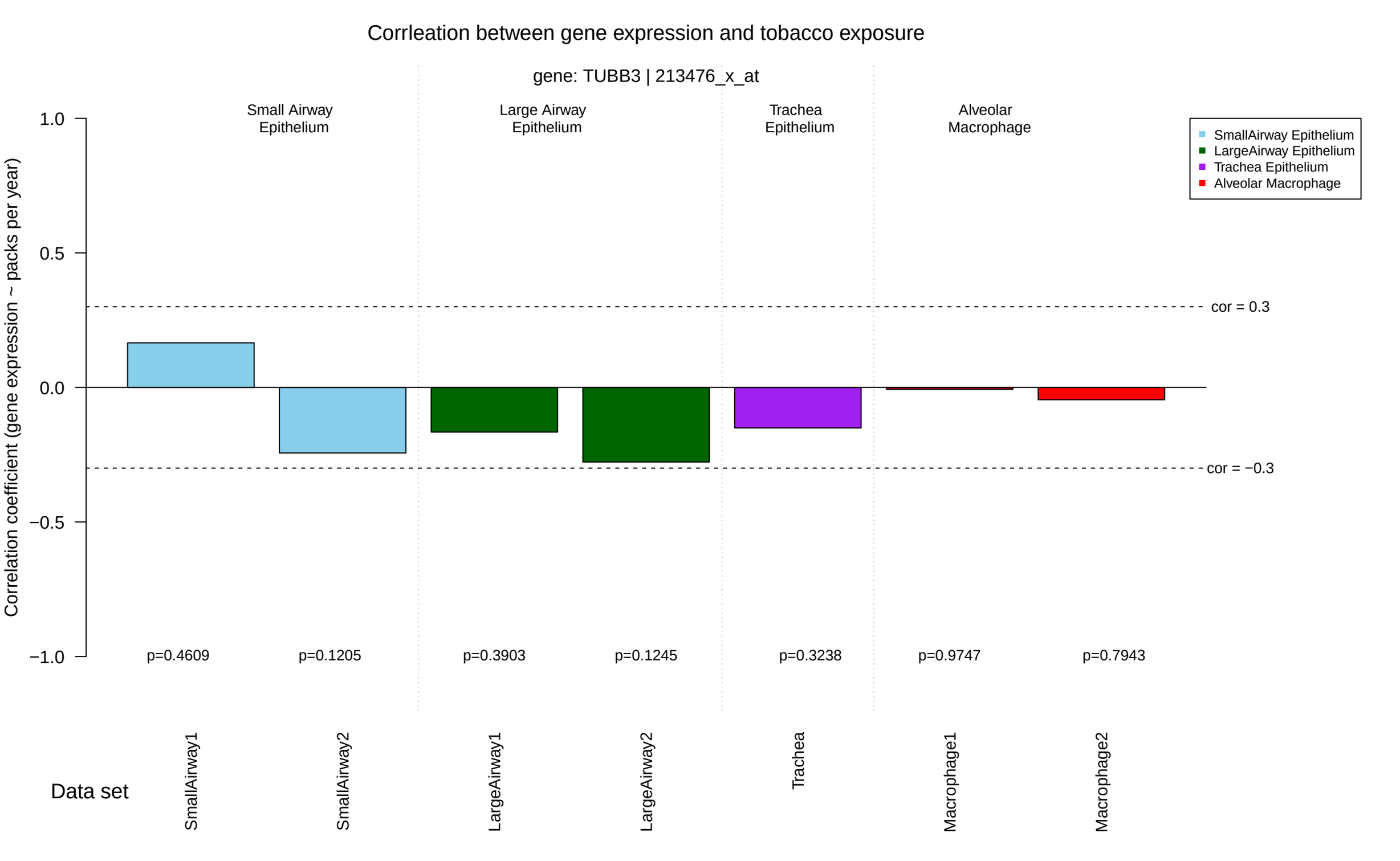

set: LargeAirway1 (p=0.2222)
TUBB3 | 213476_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.909)
TUBB3 | 213476_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2462)
TUBB3 | 213476_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8065)
TUBB3 | 213476_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4596)
TUBB3 | 213476_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TUBB3 | 213476_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4609
p=0.1205
p=0.3903
p=0.1245
p=0.3238
p=0.9747
p=0.7943
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2